

SMART LIBRARY AUTOMATION SYSTEM WITH REAL-TIME BOOK TRACKING AND GPS-BASED STUDENT ENTRY

23CS64C – PRODUCT DEVELOPMENT AND PRACTICE

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BONAFIDE CERTIFICATE

This is to certify that this project report, “Academic Management System”, is the bonafide work of **SUBASH MUTHU B (2312401)**, **SUBIKSHAN M (2212110)**, **ALWIN IMMANUEL J (2312404)** who carried out the project work under my supervision.

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CHAPTER 1

1.1 ABSTRACT

The Smart Library Automation System is a full-stack web and mobile-based application designed to digitize and modernize library operations in higher-education institutions. The system integrates multiple library processes including automatic student entry using hybrid GPS technology, real-time book location tracking using RFID tags, indoor navigation using BLE beacons, digital question paper management, transaction and fine management, reservations, and advanced analytics reporting. It provides a centralized platform for administrators, librarians, and students, ensuring seamless interaction and efficient data flow across all library activities.

The frontend is developed using React and Vite, offering a responsive and user-friendly interface, while the backend is built with Node.js, Express, and MySQL to ensure scalability, performance, and secure data handling. Role-based access control using JWT authentication enhances system security and ensures that each user interacts only with relevant functionalities.

The system eliminates manual entry registers, reduces time wasted in book searching, and provides real-time shelf tracking that is not present in conventional library management systems. Advanced features like RFID-based dynamic book location updates, BLE beacon-based indoor navigation, and automated reports further enhance usability. Overall, the solution provides a reliable, efficient, and scalable approach to managing library operations, enabling institutions to make data-driven decisions and improve student experience.

1.2 XLSCOUT REPORT

A preliminary novelty and literature exploration indicates that most existing library management solutions focus on isolated functionalities such as book cataloging, basic issue/return tracking, or simple student login systems. Traditional Library Management Systems (LMS) often store only static shelf data — meaning the system records where a

book *should* be, not where it *actually is*. This leads to the common problem of misplaced books being untrackable.

The proposed Smart Library Automation System differentiates itself by integrating multiple innovative technologies into a single unified platform. It combines hybrid GPS-based automatic student entry (using GPS, Wi-Fi, and motion sensors together), real-time RFID-based dynamic shelf tracking, BLE beacon-based indoor navigation, digital question paper storage, and live analytics dashboards within a centralized system.

The core novelty of this project lies in the Real-Time Book Location Tracking module. Unlike existing systems that maintain fixed shelf data, this system dynamically updates a book's actual physical location whenever an RFID scanner on any shelf detects it. This means even misplaced books can be found instantly. Combined with BLE beacon-based indoor step-by-step navigation, students can be guided directly to the shelf where the book currently sits — a capability absent in all conventional library systems.

Additionally, the hybrid GPS entry system avoids false check-ins by combining three signals: GPS for proximity, Wi-Fi for building confirmation, and motion sensor for walking confirmation. This multi-layered verification ensures accurate student entry records without any manual intervention. The system also supports automated peak-hour reports, department-wise usage analytics, and a digital previous-year question paper library accessible to all students.

1.3 ATTORNEY REVIEW SUGGESTION

The attorney review recommends enhancing the system by combining the physical tracking layer (RFID, BLE beacons) with a stronger software intelligence layer to enable predictive book recommendations and automated restocking alerts. It emphasizes the incorporation of advanced algorithms for tasks such as student entry confidence scoring, indoor navigation pathfinding, shelf occupancy analysis, and book demand forecasting to improve efficiency and accuracy. Furthermore, the review highlights the need to automate various manual processes including librarian report generation, overdue fine calculation, and question paper classification in order to reduce human errors and save time. By

transforming traditionally manual library workflows into intelligent and automated operations, the system can provide a more streamlined, user-friendly, and scalable solution for managing the entire library ecosystem effectively.

1.4 LITERATURE SURVEY

In recent years, higher education institutions have increasingly adopted digital systems to manage library operations efficiently. Traditional methods relying on manual registers and disconnected tools often lead to data redundancy, errors, and lack of transparency. To overcome these issues, several researchers have proposed integrated library automation solutions.

In [1], Sharma et al. developed a web-based Library Management System that centralized book records, issue tracking, and student data. Their system improved data accessibility but lacked real-time shelf tracking and indoor navigation capabilities.

In [2], Kumar and Singh proposed an RFID-based book tracking system for academic libraries. The system focused on detecting book presence at fixed reader points; however, it did not support dynamic shelf updates or BLE-based indoor guidance for students.

In [3], Reddy et al. introduced a GPS-based attendance system for classrooms. This approach improved entry accuracy but was limited to outdoor GPS and did not address the indoor accuracy problem using hybrid methods combining Wi-Fi and motion sensors.

In [4], Patel et al. developed an indoor navigation system using BLE beacons for large buildings. While it demonstrated effective zone detection, it lacked integration with library book tracking or student entry systems.

In [5], Gupta et al. proposed integration of digital repositories for previous year question papers in academic institutions. However, their model required manual uploading by administrators and lacked a student-facing search interface integrated with the main library system.

From the literature, it is evident that existing systems address specific problems such as book cataloging, RFID tracking, or indoor navigation individually. However, there is a need for a unified system that integrates all library processes including automatic entry, real-time shelf tracking, BLE navigation, and digital resource access. The proposed Smart Library Automation System aims to bridge this gap by providing a comprehensive, scalable, and secure platform for managing end-to-end library operations.

1.5 ADDITIONAL FEATURES

The Smart Library Automation System incorporates several advanced features to enhance efficiency, automation, and data integrity in library operations. The system includes a Hybrid GPS Entry module that enables students to enter and exit the library automatically without writing in manual registers, using a three-layer verification system of GPS, Wi-Fi, and motion sensors to ensure accuracy.

The core innovation is the Real-Time Book Location Tracking system using RFID tags on every book and smart scanners on every shelf. The database is updated automatically whenever a book is placed on any shelf, including wrong shelves, making the system truly dynamic. This is complemented by BLE Beacon-based Indoor Navigation that guides students step-by-step to the shelf where their required book currently sits.

The system also provides a Digital Question Paper Library where all previous year papers are stored digitally and are searchable and downloadable by students. Role-Based Access Control is implemented to maintain security, granting different access levels to administrators, librarians, and students. Advanced transaction management supports book borrowing, returns, reservations, and fine calculations with automated reminders.

Furthermore, the system supports Google OAuth login for students and a librarian dashboard with live reports on most borrowed books, peak hours, and department-wise usage. Built on a scalable full-stack architecture using React, Node.js, Express, and MySQL, the system ensures high performance, maintainability, and future scalability.

CHAPTER 2

2.1 TECHNICAL DOMAIN

The Smart Library Automation System belongs to the domain of Library Information Systems and IoT-integrated Academic Infrastructure. It focuses on automating and integrating library operations such as student entry management, real-time book tracking, indoor navigation, transaction management, and performance analytics. The system uses a full-stack web architecture with React for the frontend and Node.js with Express for the backend, supported by a MySQL database. Physical-layer technologies including RFID readers, BLE beacons, and GPS/Wi-Fi/motion sensor fusion are integrated through the backend API. It ensures secure, role-based access and efficient data handling, enabling seamless interaction between administrators, librarians, and students while supporting data-driven library decision-making.

2.2 Existing Industrial problem

In many higher-education institutions, library operations are still managed using manual processes or basic disconnected software systems such as spreadsheets, barcode scanners, and paper-based entry registers.

This leads to several challenges:

- **Manual entry bottlenecks:** Students must physically write their details in a register every time they enter or exit, creating queues during peak hours.
- **Static book location data:** Existing library software stores only the assigned shelf of a book, not its real-time physical location. If a book is misplaced, it cannot be found.
- **No indoor guidance:** Students must walk across the entire library or ask librarians repeatedly to find books.

- **Inaccessible question papers:** Previous year question papers are stored in physical cupboards with no proper organization, making it difficult for students to access them.
- **Lack of analytics:** Librarians have no digital tools to understand peak hours, most borrowed books, or department-wise usage.
- **Delayed report generation:** Usage reports and overdue tracking require significant manual effort and time.

These problems reduce operational efficiency and negatively affect both student experience and library management quality.

2.2.1 Existing Solutions

Currently, educational institutions rely on a combination of traditional and modern systems to manage library activities. These solutions can be broadly categorized into manual systems, standalone library software, ERP systems with partial implementation, and digital catalog platforms.

Manual Systems

Many institutions continue to depend on manual methods such as paper-based entry registers, Excel sheets for book records, and physical cupboards for question papers. Although these methods are simple to implement, they are highly time-consuming and prone to human errors. Additionally, such systems are not scalable and become increasingly difficult to manage as student numbers and book collections grow.

Standalone Library Software

Some colleges utilize basic library management software that supports functionalities such as book cataloging, issue and return tracking, and member management. However, these tools suffer from limited functionality and lack integration between physical tracking and digital systems. Moreover, they do not support real-time shelf tracking, indoor navigation, or hybrid GPS-based entry, restricting their effectiveness in modern library environments.

ERP Systems

Certain institutions adopt enterprise-level ERP systems such as SAP ERP and Oracle PeopleSoft to manage library and administrative processes. While these systems offer robust capabilities, they are often expensive to implement and maintain. Additionally, they are not specifically designed for IoT-based library operations such as RFID shelf tracking or BLE navigation, limiting their practical applicability in library automation contexts.

Digital Catalog Platforms

Institutions also make use of external digital catalog platforms and OPAC (Online Public Access Catalog) systems for book searching. However, these platforms show only the assigned location of a book, not its real-time position. If the book is misplaced, the catalog still shows the original shelf, leading students to a wrong location.

2.1.2 Products, Patents, and Publications

Existing Products

Several existing systems are widely used in library environments. Koha is an open-source Integrated Library System (ILS) that supports cataloging, circulation, and OPAC, making it a popular choice among institutions. Ex Libris Alma is another widely used library management platform that provides advanced analytics and cloud-based operations. RFID Library Systems by vendors such as 3M and Bibliotheca offer physical book detection at fixed points, while Springshare LibCal handles room and resource reservations.

Research Publications

Numerous research studies have been conducted in the field of library automation. The study titled "RFID-Based Library Management System for Academic Institutions" focuses on using RFID for book tracking and automated checkout. Another research work, "Indoor Navigation Using BLE Beacons in Large Public Buildings," discusses methodologies for zone detection and step-by-step routing. Additionally, the publication "Hybrid Positioning Systems for Smart Building Entry Using GPS, Wi-Fi, and Motion Sensors" explores multi-layer location verification to improve entry accuracy.

Patent Concept (General Reference)

The patent concept titled "Automated Library Book Tracking and Navigation System Using RFID and BLE" describes an integrated approach to dynamically tracking book positions and guiding users to physical shelf locations. This concept emphasizes real-time database synchronization between shelf scanners and the central system.

2.1.3 Objectives of the Proposed System

Primary Objective

- To develop a comprehensive Smart Library Automation System that integrates automatic student entry, real-time book tracking, indoor navigation, and digital resource access into a single unified platform, improving efficiency, accuracy, and student experience.

Detailed Objectives

- To automate student entry and exit using a hybrid GPS + Wi-Fi + motion sensor system, eliminating manual registers.
- To implement real-time book location tracking using RFID tags on books and smart scanners on shelves.
- To enable dynamic shelf database updates so that even misplaced books can be found instantly.
- To guide students step-by-step to the correct shelf using BLE beacon-based indoor navigation.
- To digitize the previous year question paper library, making it searchable and downloadable.
- To provide librarians with live dashboards showing peak hours, most borrowed books, and department-wise usage.

- To automate transaction management including borrowing, returns, fine calculation, and reservations.
- To support Google OAuth login for students and role-based access for librarians and administrators.
- To generate automated reports for management decision-making.
- To build a scalable full-stack system capable of handling large book collections and student populations.

CHAPTER 3

SYSTEM ARCHITECTURE

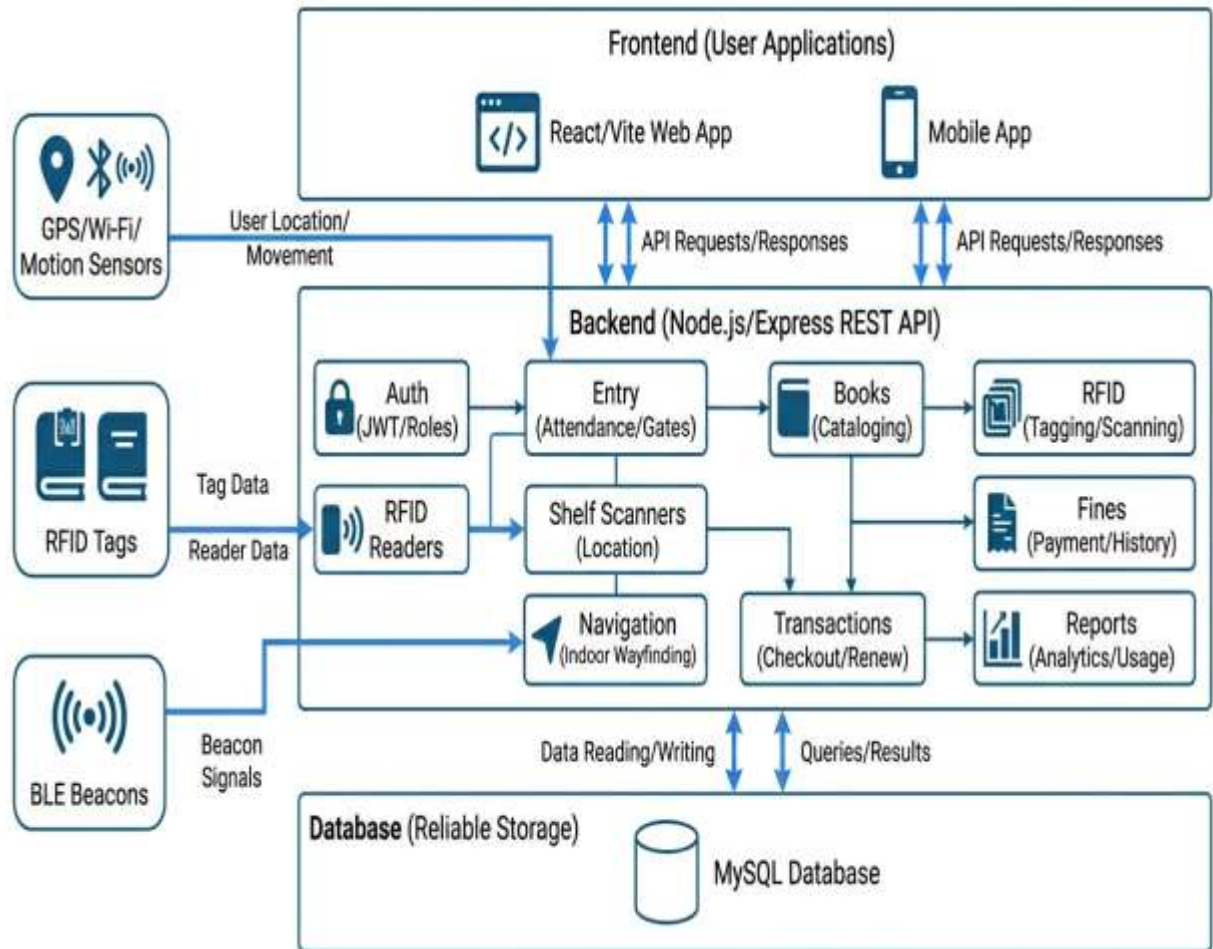


Fig.3.1 System Architecture Diagram

Figure 3.1 illustrates the overall architecture of the Smart Library Automation System showing interaction between the physical hardware layer, the backend API, and the frontend interfaces.

3.2 Admin module

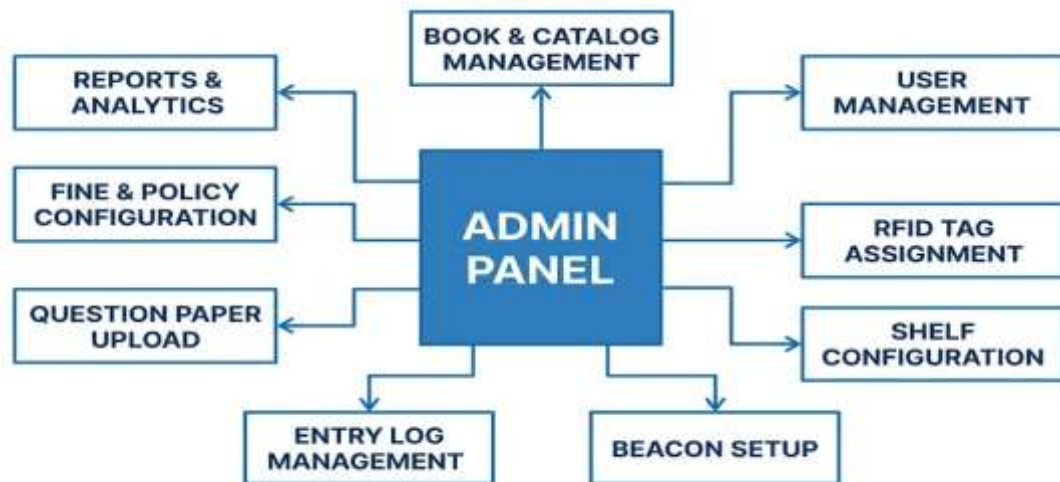


Fig 3.2 Admin module diagram

Figure 3.2 illustrates the functionalities of the Admin Module for managing the library's physical and digital infrastructure.

The Admin Module acts as the central control unit of the Smart Library Automation System and is responsible for configuring and managing all core library resources. The Book and Catalog Management component enables the administrator to add, edit, and remove books from the catalog, assign RFID tags to physical books, and link books to their designated shelves. The Shelf Configuration module allows creation and management of shelf zones, floors, and sections, which form the physical map used by the navigation system.

The Beacon Setup component facilitates placement and registration of BLE beacons near shelf zones so that the indoor navigation engine can guide students accurately. The User Management module handles student and librarian accounts, role assignments, and access permissions. The Entry Log Management component allows the administrator to review entry and exit records, override incorrect entries, and audit hybrid GPS confidence scores.

The Question Paper Upload section provides tools to scan, upload, and categorize previous year question papers by department, subject, and year. The Fine and Policy Configuration module allows setting borrowing durations, fine rates per day, and reservation policies. The

Reports and Analytics module provides comprehensive insights into library usage, including peak hours, most borrowed books, department-wise data, and student attendance patterns.

3.3 Librarian module

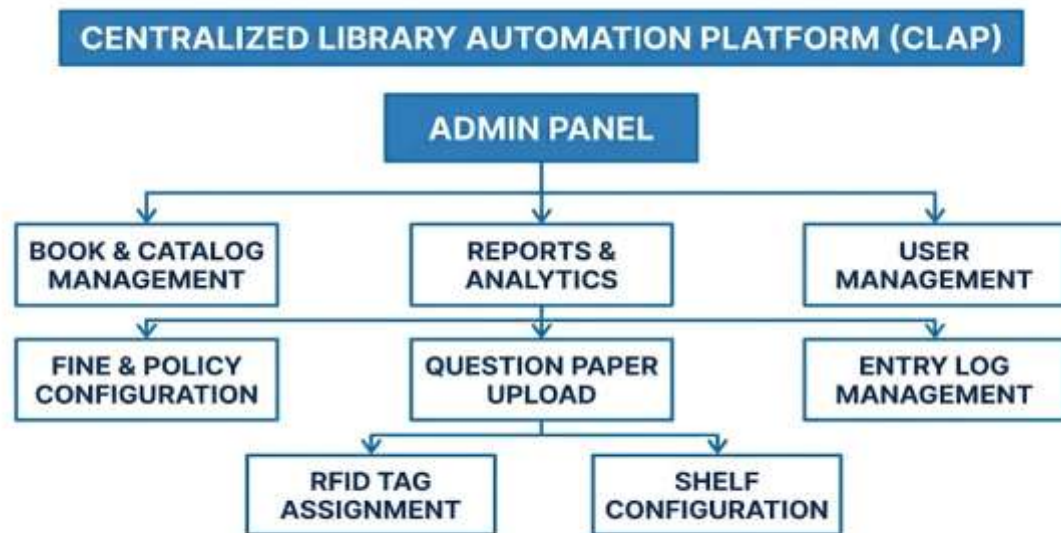


Fig 3.3 Staff Module Diagram

Figure 3.3 illustrates the Librarian Module for handling day-to-day book transactions, RFID monitoring, and student management.

The Librarian Module represents the functionalities available to library staff for managing daily library operations. The Librarian Dashboard serves as the central interface, providing access to all relevant library operations. The Book Issue and Return module enables librarians to process borrowing and return transactions by scanning RFID tags or searching by book name, updating the database instantly.

The Real-Time Book Location View component allows librarians to search for any book and immediately see its current shelf location as detected by the RFID shelf scanner, including cases where the book has been misplaced. The RFID Scanner Management module allows librarians to manage handheld RFID readers and monitor fixed shelf reader status. The Reservation Management component handles student book reservation requests, approving or rejecting them based on availability.

The Fine Collection module displays overdue books, calculates outstanding fines automatically, and records fine payments. The Overdue Alerts component sends automated notifications to students with overdue books. The Report Generation section provides one-click reports on daily transactions, active borrowings, and fine collections.

3.4 Student Diagram

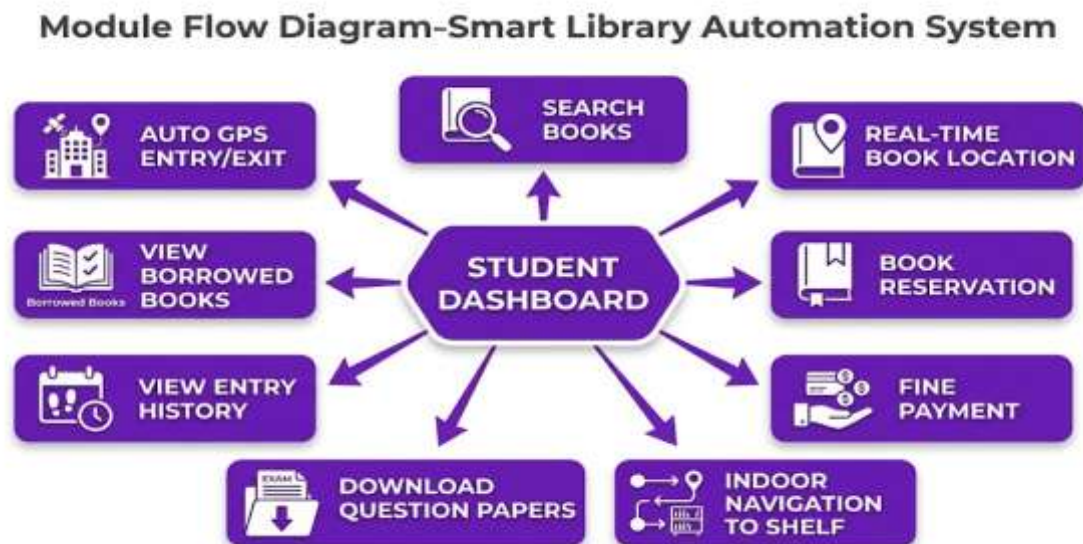


Fig 3.4 Student Module Diagram

Figure 3.4 illustrates the Student Module, which provides access to book search, navigation, borrowing history, and digital resources.

The Student Module represents the functionalities available to students for accessing library resources and performing self-service operations. The Student Dashboard acts as the central interface, connecting students to all library features. The Auto GPS Entry/Exit component handles automatic library check-in and check-out using the hybrid GPS system, requiring no manual action from the student.

The Search Books module allows students to search the catalog by title, author, subject, or department and immediately see the current real-time shelf location of the book as updated by the RFID system. The Indoor Navigation to Shelf component uses BLE beacon signals detected by the student's phone to provide step-by-step walking directions to the exact shelf

where the book currently sits. The View Borrowed Books section shows currently borrowed books, due dates, and renewal options.

The Book Reservation module enables students to reserve books that are currently issued to others, placing them in a queue. The Fine Payment component shows outstanding fines and supports online payment. The Download Question Papers section provides a searchable digital repository of previous year question papers, downloadable by subject and year.

3.5 Entry Management module

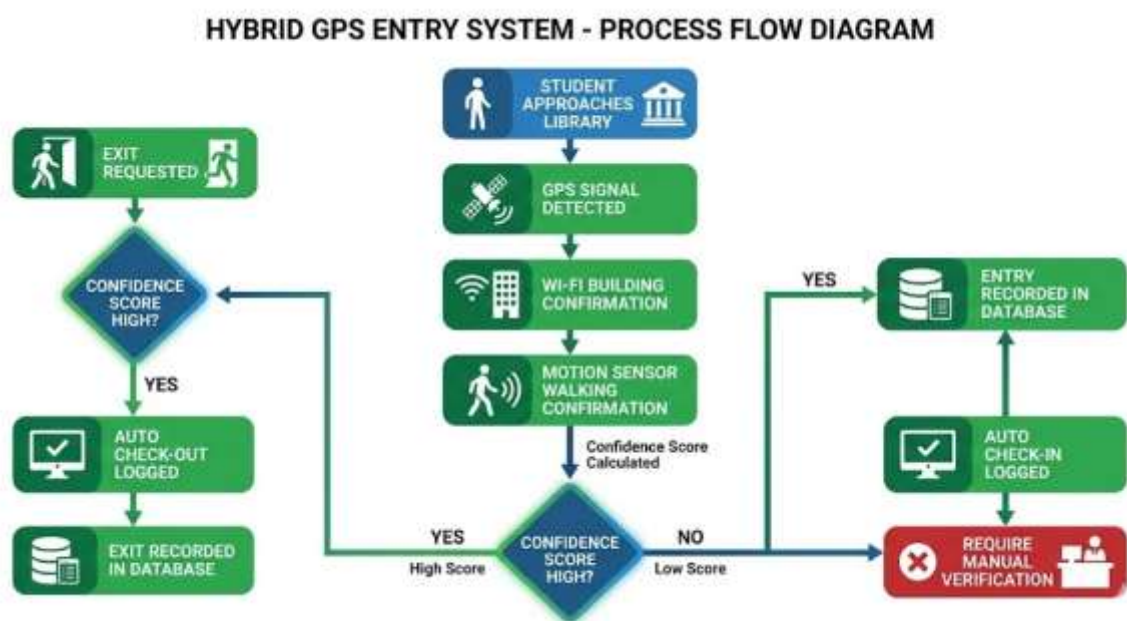


Fig 3.5 Attendance Module Diagram

Figure 3.5 illustrates the process of automatic student entry detection and verification using the hybrid GPS, Wi-Fi, and motion sensor system.

The Entry Management Module is a key innovation of the Smart Library Automation System that ensures accurate and fully automated tracking of student entry and exit without any manual action. The process begins when a student approaches the library building and their mobile device detects GPS coordinates near the library location. The system then verifies that the student's device is connected to the library's Wi-Fi network, confirming they are inside the building rather than just nearby. A motion sensor check then confirms

that the student is actively walking, distinguishing an entering student from one sitting outside.

These three signals are combined into a confidence score. If the score exceeds the threshold, the system logs a check-in entry automatically, recording the timestamp, student ID, GPS coordinates, Wi-Fi confirmation status, and motion confidence. The database entry_logs table stores all these details for audit and reporting purposes. The same multi-layer process works in reverse for check-out when the student leaves the library. This eliminates queues, prevents false entries, and provides accurate real-time occupancy data to the librarian dashboard.

3.6 Real-Time RFID Book Tracking Module



Fig 3.6 Timetable and Staff Allocation Module Diagram

Figure 3.6 illustrates the real-time RFID-based book tracking process that dynamically updates book shelf locations in the database whenever a book is placed on any shelf.

The RFID Book Tracking Module is the core novelty of the Smart Library Automation System. Every physical book in the library is attached with a unique RFID tag that carries a tag ID linked to the book's catalog entry in the database. Every shelf in the library is

equipped with a fixed RFID scanner that continuously monitors which tags are present on that shelf.

When a book is placed on any shelf — whether the correct shelf or a wrong one — the shelf scanner reads the RFID tag, identifies the shelf ID, and sends an automatic update to the backend API. The database immediately updates the book's current shelf location to the shelf where it was just detected. This means the system always knows the real, physical, current location of every book — not just the originally assigned shelf.

This is fundamentally different from traditional library software, which stores only a fixed "home shelf" for each book and has no way of knowing if the book has been misplaced. When a student searches for a book, the system shows the actual shelf where the book currently sits. Even if a librarian placed the book on the wrong shelf, the system accurately reports its real position, saving students significant time.

3.7 BLE Beacon Indoor Navigation Module

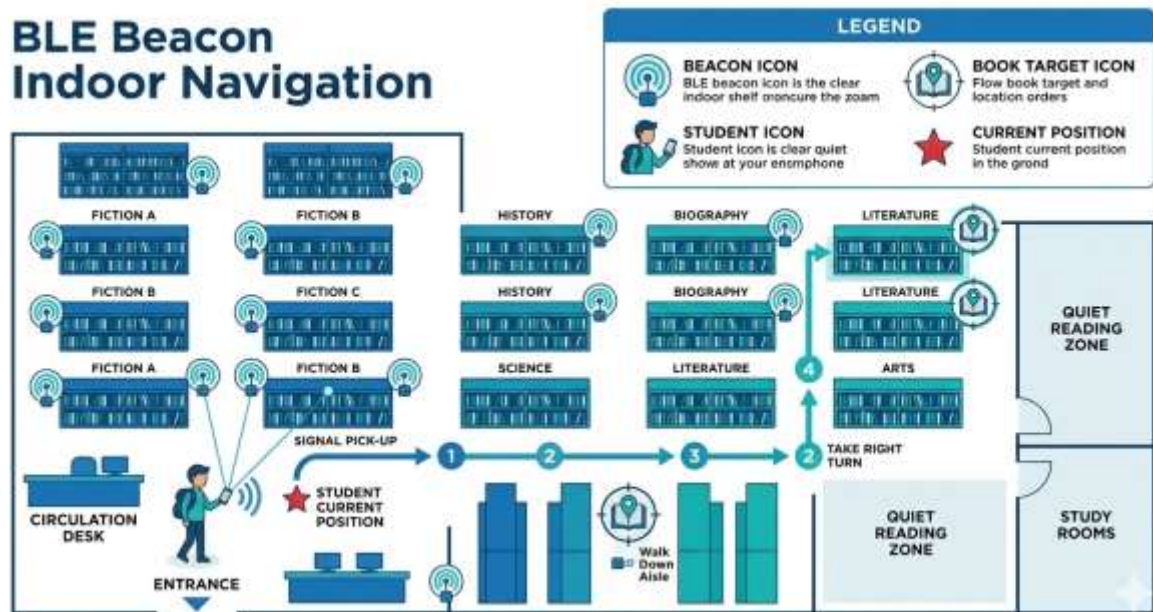


Fig 3.7 CO Based Module Diagram

Figure 3.7 illustrates the BLE beacon-based indoor navigation system that guides students step-by-step to the shelf where their required book is currently located.

The Indoor Navigation Module addresses the well-known limitation of GPS: it does not work accurately inside buildings. To solve this, BLE (Bluetooth Low Energy) beacons are placed near each shelf zone throughout the library. When a student searches for a book and the system identifies the current shelf from the RFID tracking database, the navigation module calculates a step-by-step path from the student's current position to the target shelf.

The student's phone detects signals from nearby BLE beacons and determines which zone they are currently in. The system then provides turn-by-turn walking directions — for example, "Go straight past Shelf A3, turn left at Shelf B1, your book is on Shelf B4 on the right side."

CHAPTER 4

4.1 Methodology FlowChart

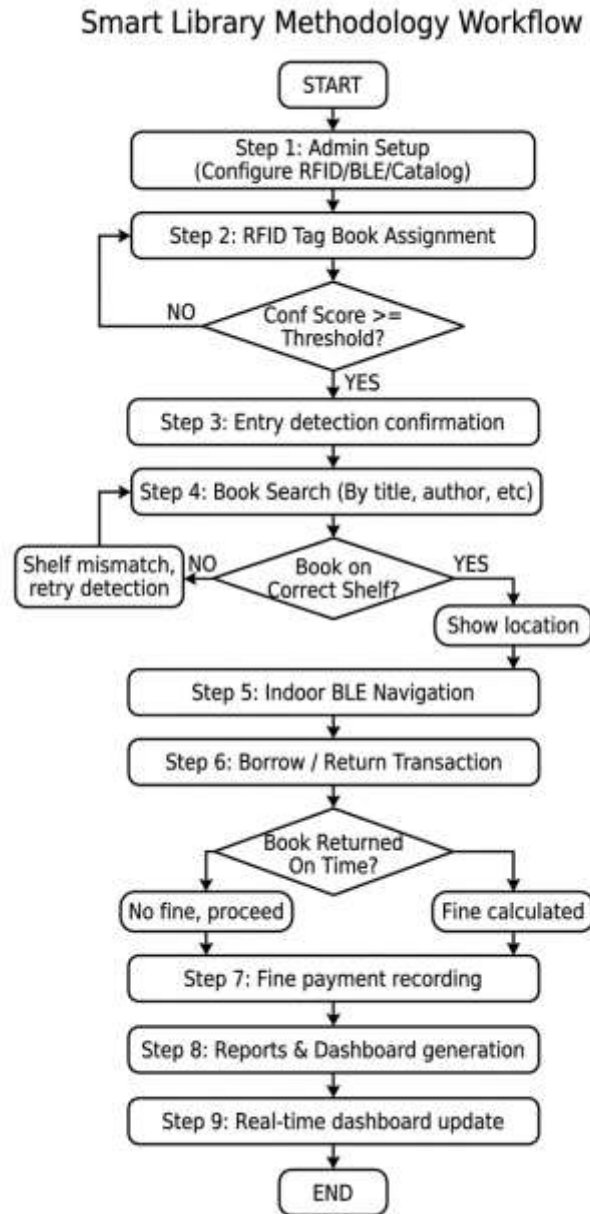


Fig 4.1 Methodology FlowChart

Figure 4.1 illustrates the complete workflow of the system from initial setup through student entry, book discovery, borrowing, and report generation.

4.2 Overview of the System Workflow

The Smart Library Automation System follows a systematic and modular workflow to automate and integrate all library activities. The methodology is designed to ensure smooth coordination between administrators, librarians, and students. It begins with system configuration, followed by RFID tag assignment, student entry, book searching, indoor navigation, transaction processing, fine management, and reporting. Each stage is interconnected and ensures data consistency, accuracy, and real-time synchronization throughout the library lifecycle.

4.3 Administrative Configuration

The first phase of the system involves administrative setup, which forms the foundation of all library operations. The administrator is responsible for configuring key entities such as shelf zones, floor plans, RFID reader positions, BLE beacon placements, and library policies. These elements define both the physical map of the library and the digital structure of the system.

Once the physical map is defined, the system allows the administrator to register all books in the catalog, assigning each physical book a unique RFID tag linked to its catalog entry. Following this, RFID shelf scanners are registered for each shelf, and BLE beacons are registered for each shelf zone with their physical coordinates. The administrative configuration phase ensures that all hardware and software components are connected and ready before students begin using the system.

4.4 Student Entry and Exit Process

After system setup, students interact with the library through the automatic entry system. When a student approaches the library, their mobile device begins the hybrid verification process. The GPS module checks proximity to the library's GPS coordinates. The Wi-Fi module confirms connection to the library's network. The motion sensor module confirms active walking. These three signals are combined into a confidence score, and if it exceeds the threshold, the entry is automatically logged without any action from the student.

The system records the student's ID, timestamp, GPS coordinates, Wi-Fi status, motion confidence score, and entry type (entry or exit) in the entry_logs table. This data feeds into the librarian's real-time occupancy dashboard, showing how many students are currently inside the library and when peak hours occur.

4.5 Book Search and Real-Time Location Retrieval

Once inside, students use the search interface to find a book by title, author, subject, or department. The system queries the book catalog and returns matching results. For each result, the system fetches the book's current shelf location from the RFID tracking database — this is the last shelf where the RFID scanner detected the book's tag, which may differ from the book's originally assigned shelf if it was misplaced.

The student sees the actual current shelf of the book, not just its assigned location. This real-time location data is what makes the system fundamentally different from traditional catalog systems. If multiple copies exist, the system shows the current shelf of each copy individually.

4.6 Indoor Navigation to Book Shelf

After the student identifies the book and its current shelf, they can activate indoor navigation. The system calculates a step-by-step path from the student's current beacon zone to the beacon zone near the target shelf. As the student's phone continuously scans for BLE beacon signals, the system detects which zone they are in and updates the navigation instructions accordingly.

The navigation instructions are displayed as simple text directions on the student's mobile screen, for example: "You are at Zone A. Walk forward 10 meters to Zone B. Turn right. Your book is on Shelf B4." This process continues until the student reaches the correct shelf, at which point the system confirms arrival and highlights the exact shelf section where the book sits.

4.7 Book Borrowing and Return Transaction

When a student wishes to borrow a book, the librarian scans the book's RFID tag using a handheld reader. The system links the book to the student's account, records the issue date, and sets the due date based on library policy. The book's status in the database is updated to "issued," and its shelf location is cleared until the book is returned.

When the student returns the book, the librarian scans it again. The system closes the transaction, calculates any outstanding fines if the return is overdue, and records the return timestamp. When the librarian places the returned book on a shelf, the RFID shelf scanner automatically detects it and updates the database with the new shelf location in real time.

4.8 Fine Calculation and Payment

The system automatically calculates fines for overdue books based on the configured fine rate per day. Students can view their outstanding fines on the student dashboard. Fine payment can be processed by the librarian and recorded in the payments table with a receipt generated automatically. The fine management module maintains a complete history of all fines issued and payments received.

4.9 Digital Question Paper Access

Students can access the question paper library from their dashboard without visiting any physical cupboard. The system provides a search interface where students can filter question papers by department, subject, year, and semester. Papers are stored digitally on the server and can be downloaded as PDF files. New question papers can be uploaded by administrators through the admin panel with proper categorization.

4.10 Reporting and Analytics

The final stage of the methodology involves report generation and dashboard analytics. The librarian dashboard provides live reports including current occupancy count, most borrowed books, peak hour graphs, department-wise borrowing statistics, and overdue book alerts.

CHAPTER 5

5 System Testing and Result

System testing is conducted to ensure that the Smart Library Automation System operates correctly, satisfies all specified requirements, and remains free from critical errors. As the system is developed as a full-stack web application with physical hardware integration, testing is performed at multiple levels to validate both individual software components and hardware-software interactions. This comprehensive testing approach ensures reliability, security, and accuracy across all modules.

5.1 Validation Testing

Validation testing is performed to ensure that only correct and meaningful data is entered into the system. Input fields for book catalog entries, student registration, shelf configurations, and fine policies are checked for required values, appropriate data types, and acceptable formats. Database constraints are also used to prevent duplicate RFID tag assignments and ensure each tag is linked to exactly one book. The testing results confirm that invalid inputs are effectively rejected and appropriate error messages are displayed, ensuring high data quality and system reliability.

5.2 Authentication and Security Testing

Security testing is carried out to verify that the system protects user data and restricts unauthorized access. JWT-based authentication is tested to ensure secure session management, while Google OAuth integration is validated for student login. Password encryption using bcrypt is validated for librarian and admin accounts. Role-based access control mechanisms are checked to confirm that students cannot access librarian functions, and librarians cannot access admin configuration panels. API endpoints are also secured against unauthorized usage. The results demonstrate that the system successfully enforces security measures and protects sensitive information.

5.3 RFID and Hardware Integration Testing

Hardware integration testing is conducted to verify that RFID shelf scanners correctly detect book tags and transmit location updates to the backend in real time. Test books with RFID tags are placed on various shelves including intentionally wrong shelves, and the system is verified to correctly report the actual detected shelf in all cases. BLE beacon zone detection is tested by walking with a mobile device through different library zones and verifying that the correct zone is identified at each position. Entry detection is tested by simulating student approach scenarios with varying GPS signal strengths and Wi-Fi conditions.

5.4 Database Testing

Database testing is conducted to verify data consistency, integrity, and proper relationships between tables. Constraints such as primary keys, foreign keys between books and RFID tags, and unique constraints on tag IDs are validated. Transaction handling is tested to confirm that book issue and return operations are atomic, with no partial updates left in the database in case of failures. The RFID location update pipeline is stress-tested with simultaneous updates from multiple shelf scanners to ensure no race conditions occur.

5.5 Possible Future Enhancements

The Smart Library Automation System can be further improved by incorporating advanced features to enhance usability, performance, and scalability. One potential enhancement is the development of a dedicated mobile application with improved BLE scanning capabilities for more precise indoor navigation. The integration of AI-based book recommendation systems can provide personalized suggestions to students based on their borrowing history and academic department.

CHAPTER 6

INTERPRETATION AND RESULTS

6.1 Smart Library Automation System Project Flow

Tech Stack

- Frontend: React.js
- Frontend Styling: Tailwind CSS
- Frontend HTTP Client: Axios
- Frontend Icons: Lucide React
- Backend Runtime: Node.js
- Backend Framework: Express.js
- Backend Environment Config: dotenv
- Database: MySQL 8.0+
- Database Driver: mysql2
- Authentication: JWT (JSON Web Tokens) + Google OAuth
- Session Management: express-session
- Security: Helmet, CORS, express-rate-limit
- Hardware Integration: RFID reader APIs, BLE beacon scanning, GPS/Wi-Fi/Motion sensor fusion
- API Architecture: REST API under /api/v1
- Development Tools: npm, Vite, package.json

The Smart Library Automation System is a full-stack web application developed to digitalize and streamline college library activities. The system follows a three-layer architecture consisting of the frontend, backend, and database, with an additional physical hardware layer. The frontend provides role-based dashboards and user interfaces for Administrators, Librarians, and Students. The backend handles business logic, hardware data ingestion, authentication, API processing, and communication with the database. The MySQL database stores all library and user-related information in a structured format. The project flow begins with user authentication. Every user logs into the system using valid credentials via Google OAuth or their library ID. Based on the role assigned in the system, the user is redirected to the respective dashboard. The Admin manages shelves, beacons, book catalog, RFID tag assignments, and library policies. The Librarian handles daily transactions, RFID scanner operations, fine collections, and report generation. Students can search books, view real-time locations, navigate to shelves, manage borrowings, pay fines, and download question papers.

6. FOLDER STRUCTURE

6.1 BACKEND FOLDER STRUCTURE



Fig 6.1 Backend Folder Structure

Figure 6.1 illustrates the overall organization of backend files and directories showing modular architecture.

In a Node.js and Express-based application, the routes, controllers, and middleware together form the backbone of the backend architecture, ensuring modularity, readability, and maintainability. Routes define the various endpoints through which clients interact with the server. Each route file handles a specific domain: authentication, books, RFID, beacons, entry logs, navigation, transactions, fines, and reservations.

Controllers act as the link between routes and business logic, processing incoming requests, interacting with the database, and returning appropriate responses. The database folder contains the SQL schema defining all tables including books, rfid_tags, shelves, readers, beacons, entry_logs, transactions, fines, and reservations. The ensure-ready.js script handles runtime table verification and column additions to maintain compatibility across deployments.

6.2 FRONTEND FOLDER STRUCTURE

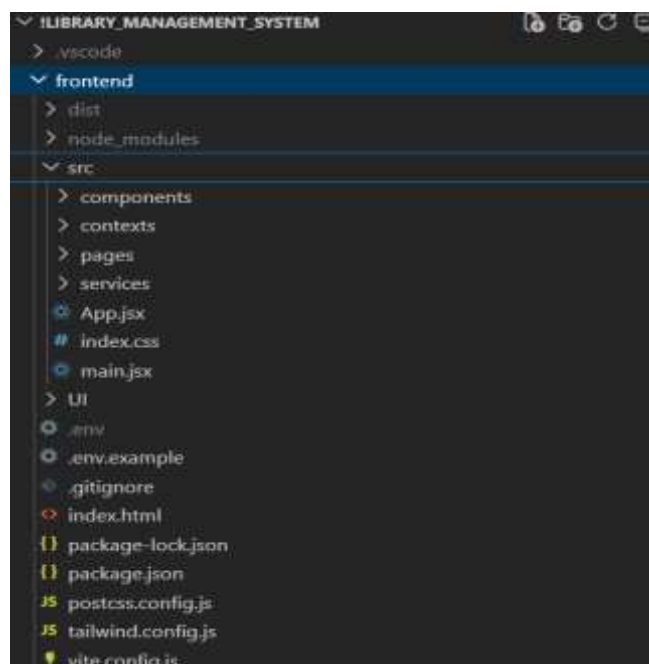


Fig 6.2 – Frontend Folder Structure

Figure 6.2 illustrates the layout of the React-based frontend directories with components and pages.

ADMIN DASHBOARD

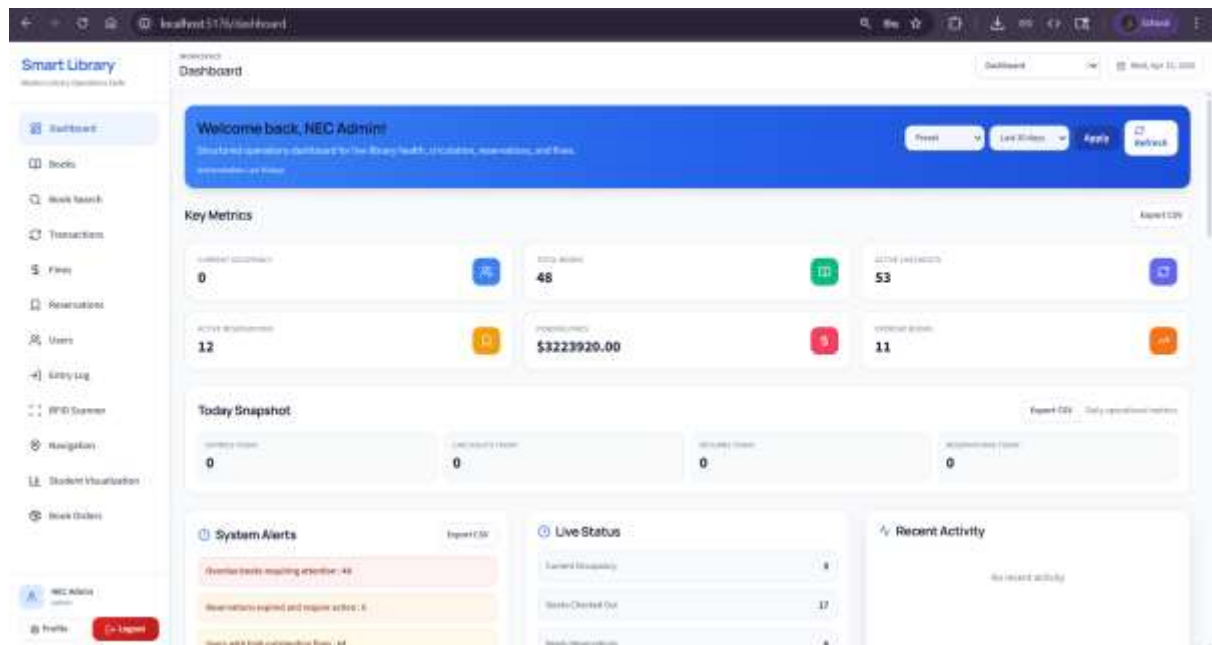


Fig 6.3 Admin Dashboard

Figure 6.3 illustrates the Admin Dashboard summarizing library metrics and providing quick access to management functions.

The Admin Dashboard is designed to help manage the library efficiently by providing a structured and user-friendly interface. The layout is divided into three main sections — the header and navigation sidebar, key metric cards, and management panels containing essential administrative functions.

At the top, the page header displays the title "Admin Dashboard" along with the subtitle "Manage your library system efficiently." On the left side, the navigation sidebar offers quick access to various modules. The central part of the page contains key metric cards providing a quick statistical summary: Total Books in catalog, number of Active Students registered, Books Issued Today, and Current Library Occupancy based on the automatic GPS entry system. These metrics allow the admin to monitor library data at a glance.

Below the metrics, the management options section presents essential action buttons that enable administrators to perform core operations such as adding books, assigning RFID

tags, configuring shelves, and uploading question papers. The Recent Transactions section lists the most recent book borrowings, helping administrators track ongoing library activity.

BOOK CATALOG MANAGEMENT

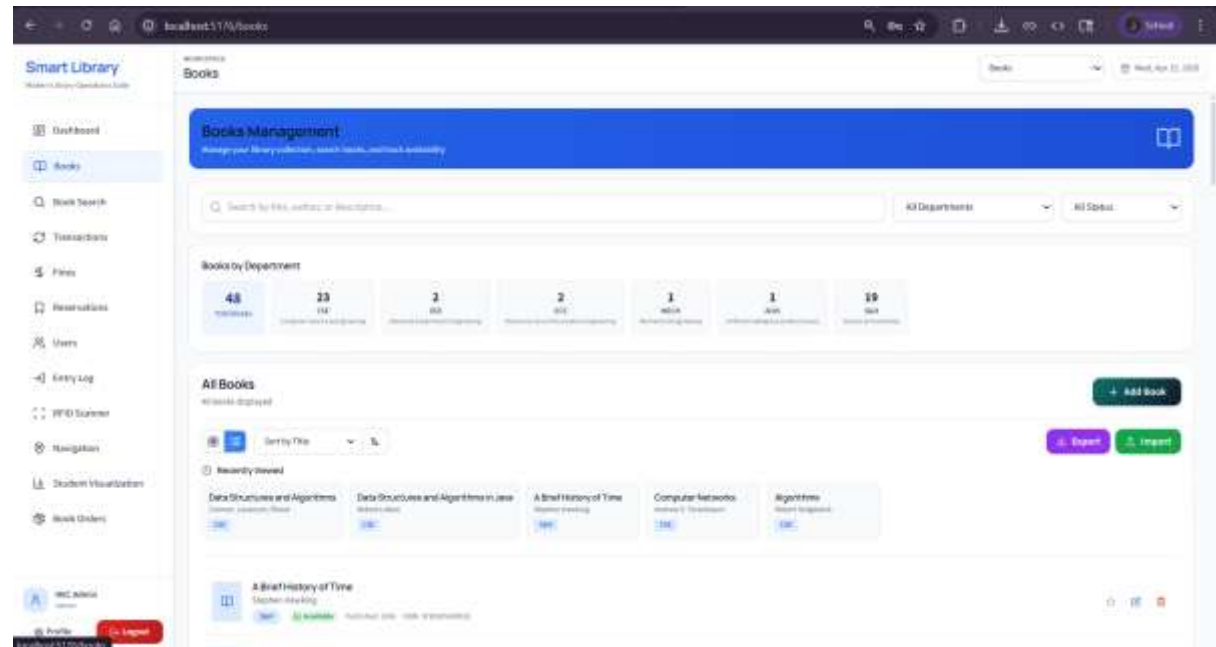


Fig 6.4 Book Catalog Management Page

Figure 6.4 illustrates the Book Catalog Management page for adding, editing, and monitoring the real-time shelf location of all books.

The Book Catalog Management page provides administrators with a comprehensive view of all books in the library. The key innovation visible on this page is the dual shelf column — Assigned Shelf shows where the book is supposed to be, while Current Shelf shows where the RFID system last detected it. When these two values differ, the row is highlighted in amber, alerting staff that the book is misplaced. Despite the mismatch, the current shelf value is always accurate and is what the student navigation system uses to guide students to the correct location.

RFID MANAGEMENT

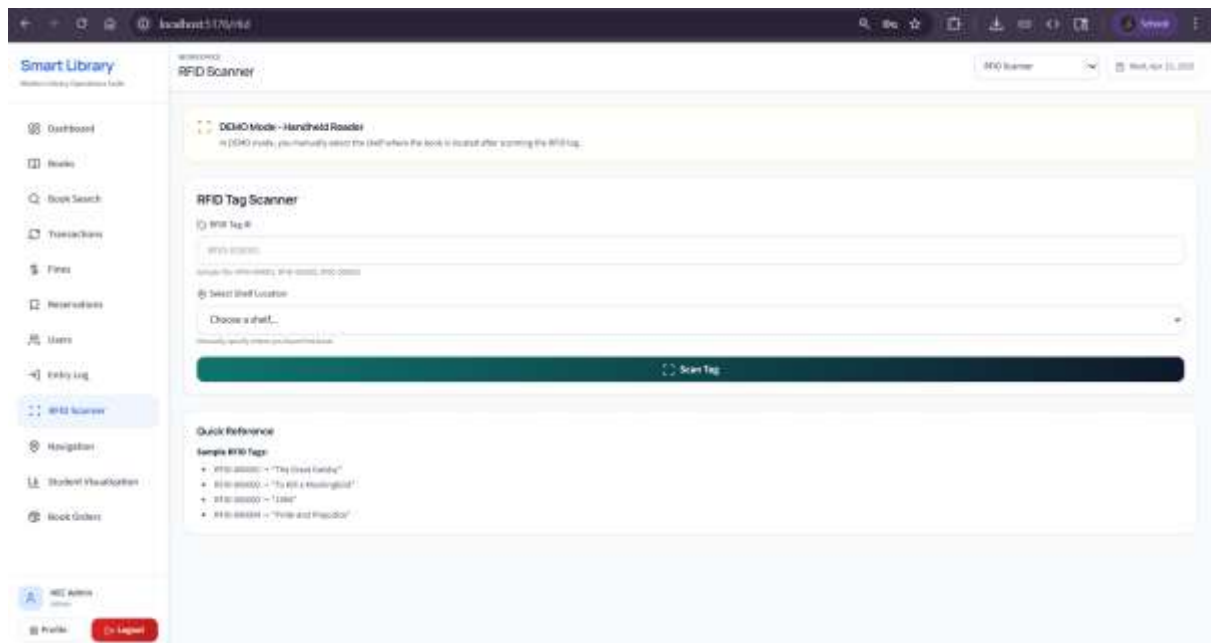


Fig 6.5 RFID Tag Assignment and Scanner Status Page

Figure 6.5 illustrates the RFID Tag Assignment and Scanner Status page for managing physical book tracking hardware.

The RFID Management page is a critical administrative interface that bridges the physical and digital layers of the system. The tag assignment section allows administrators to link a specific RFID tag ID to a book's catalog entry, establishing the digital identity of the physical book. The scanner status section provides real-time monitoring of all shelf scanners in the library. If a scanner goes offline, administrators are alerted immediately so they can investigate and restore tracking coverage for that shelf zone.

STUDENT DASHBOARD

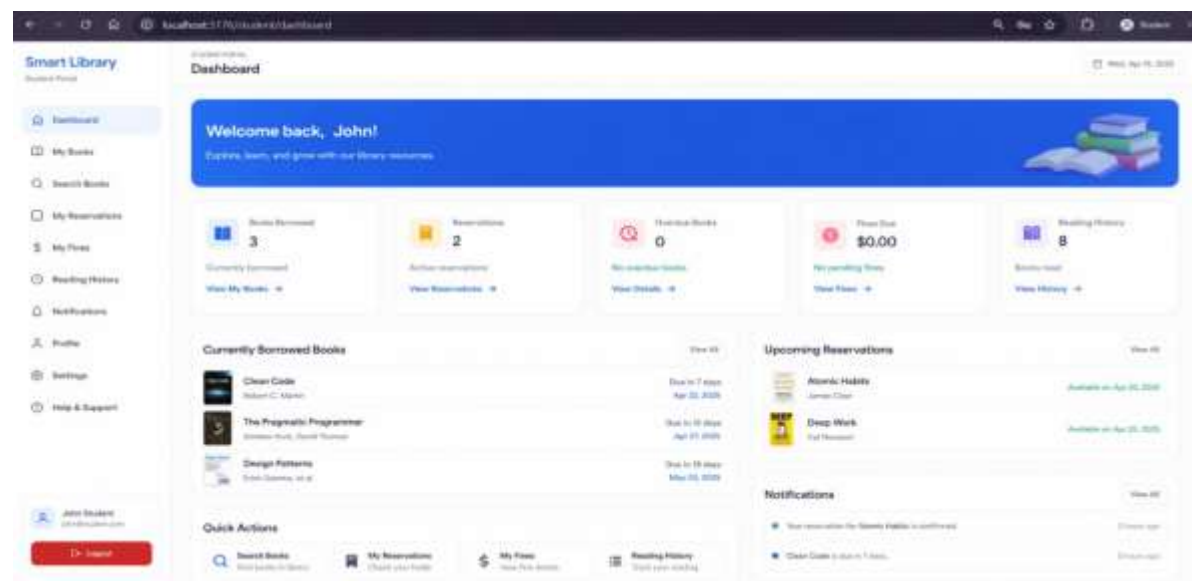


Fig 6.6 Student Dashboard

Figure 6.6 illustrates the Student Dashboard providing personalized library access with real-time book search and navigation.

The Student Dashboard provides students with a personalized, data-rich interface for interacting with the library. The entry status badge at the top shows whether the student is currently checked into the library based on the automatic GPS system, confirming that the entry system has correctly recorded their presence. The Search Books section is the most frequently used feature, allowing students to search the catalog and immediately see the current real-time shelf location of each book. The "Navigate to Shelf" button on each result card activates the BLE beacon navigation system, providing step-by-step directions from the student's current position to the book's detected shelf.

BOOK NAVIGATION INTERFACE

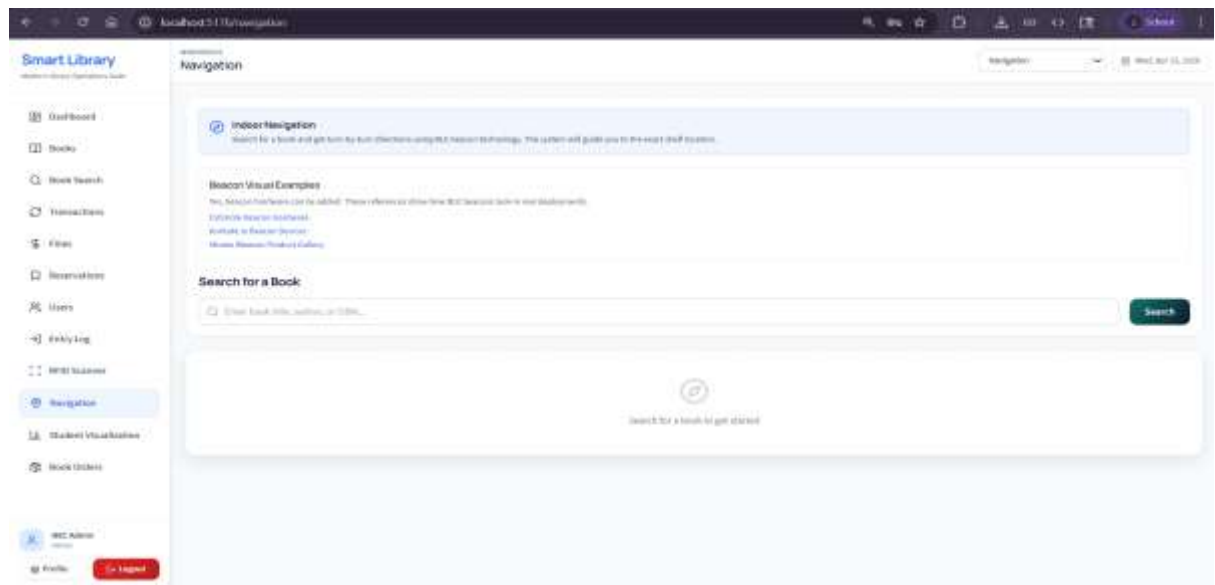


Fig 6.7 Book Navigation Interface

Figure 6.7 illustrates the Indoor Navigation Interface guiding students step-by-step to the shelf where their required book is currently located.

The Indoor Navigation Interface is the student-facing component of the BLE beacon navigation system. The floor plan view shows the student's current zone (detected from nearby beacon signals) and the target shelf (retrieved from the RFID tracking database). The dotted path represents the calculated route. As the student moves through the library, their position icon updates in real time as new beacons are detected. The text instructions below the map are updated simultaneously, always showing the remaining steps from the current position.

LIBRARIAN TRANSACTION MANAGEMENT

Smart Library

Student Library Management System

Dashboard

Books

Book Search

Transactions

Fines

Reservations

Users

Library Log

RFID System

Navigation

Student Identification

Book Status

MS Admin

Profile

Logout

Smart Library

Transactions

Transactions

Manage Book Transactions and Status

ALL

In Use

Returned

Overdue

Checklist Book

ID	USER	BOOK	CHECKOUT DATE	DATE DUE	STATUS	REMARKS
400	David Student	To All a Goodnight	04-Apr-2020	04-Apr-2020	IN USE	Return
4100	David Student	Signal Detection	05-Apr-2020	04-Apr-2020	RETURNED	
4100	Emma Schuler	The Catcher in the Rye	05-Apr-2020	04-Apr-2020	RETURNED	
000	Emma Schuler	Click & Machinery	05-Apr-2020	04-Apr-2020	IN USE	Return
4107	David Student	Data Structures and Algorithms	05-Apr-2020	02-Apr-2020	RETURNED	
4109	Emma Schuler	To All a Goodnight	05-Apr-2020	02-Apr-2020	RETURNED	
4100	Frank Reader	Efficient Java	04-Apr-2020	02-Apr-2020	RETURNED	
007	Frank Reader	Microprocessors and Interfacing	04-Apr-2020	03-Apr-2020	IN USE	Return
4100	David Student	Exquisite Body YTM	04-Apr-2020	03-Apr-2020	RETURNED	
4108	Emma Schuler	Windows Application	04-Apr-2020	02-Apr-2020	RETURNED	
000	David Student	Microprocessors and Interfacing	01-Apr-2020	02-Apr-2020	IN USE	Return

Fig 6.8 Librarian Transaction Management Page

Figure 6.8 illustrates the Librarian Transaction Management page for processing book issues, returns, and monitoring overdue books.

The Transaction Management page gives librarians complete visibility over all active borrowing transactions. The overdue rows are highlighted in red, drawing immediate attention to books that need follow-up. The Actions column provides one-click return processing — when a librarian clicks Return and scans the book's RFID tag, the system closes the transaction, calculates any fine, and waits for the RFID shelf scanner to detect the book's placement on a shelf before updating its location. This seamless handoff between the transaction system and the RFID tracking system ensures the book's location is always current.

FINE MANAGEMENT

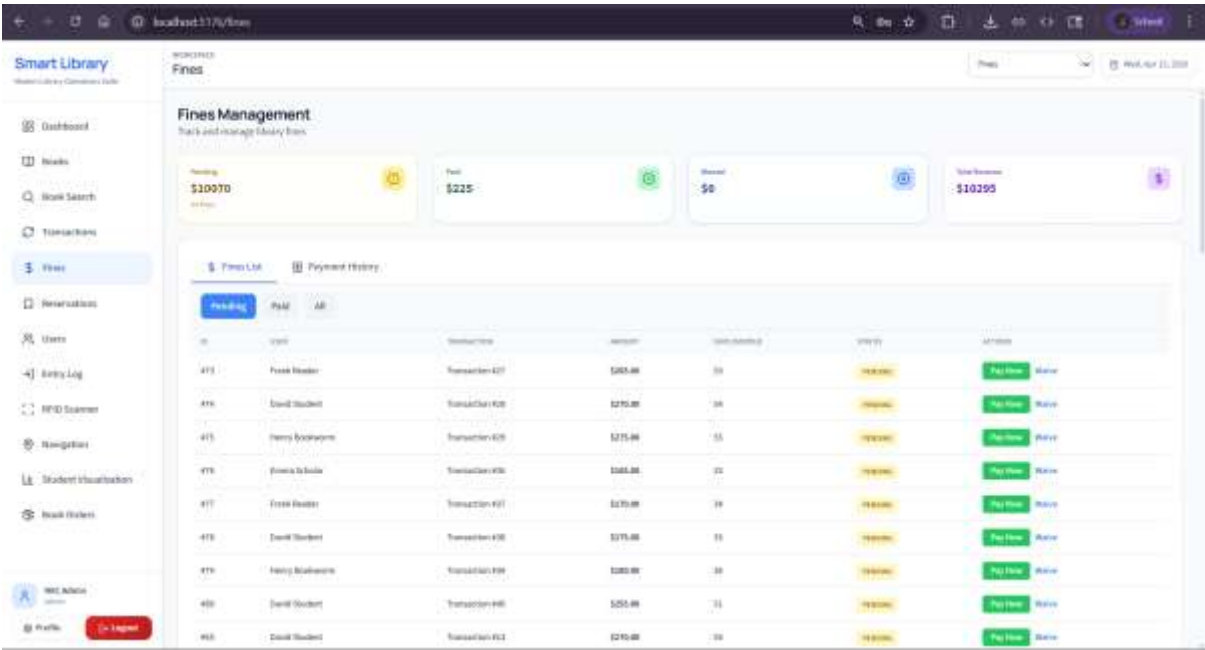


Fig 6.9 Fine Management Page

Figure 6.9 illustrates the Fine Management page for tracking, collecting, and recording library fines.

The Fine Management page provides librarians with a complete record of all fines in the system. The system automatically calculates fine amounts based on the number of overdue days multiplied by the configured daily fine rate. Librarians do not need to manually compute any amounts. When a fine is paid, the librarian marks it as paid and the system generates a digital receipt linked to the student's transaction record. The summary cards at the top provide a quick daily view of fine collection performance.

QUESTION PAPER LIBRARY

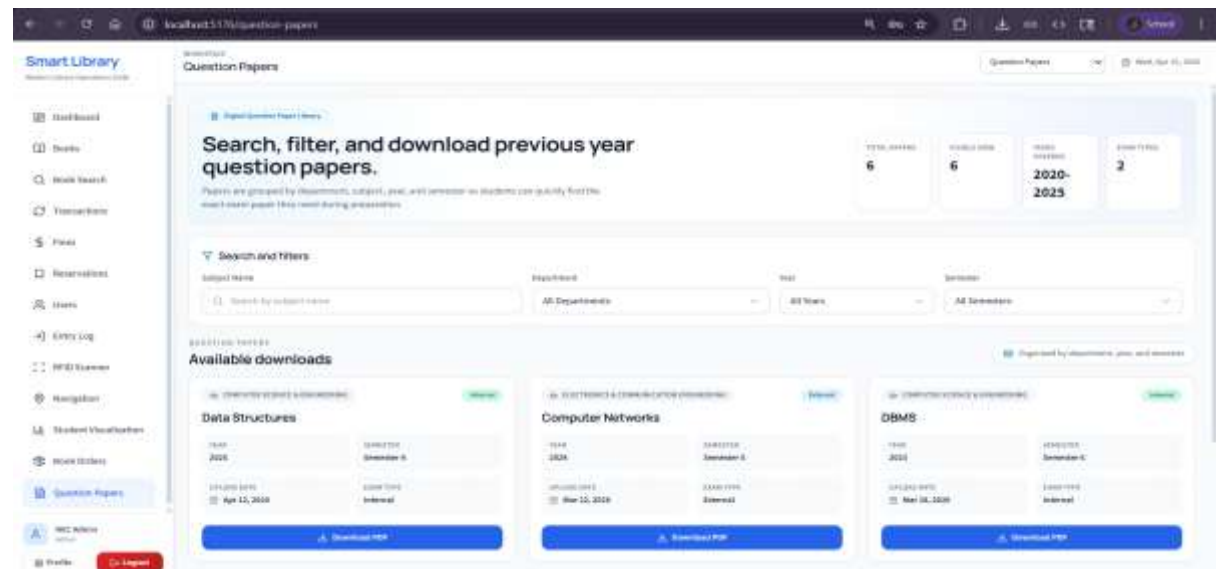


Fig 6.10 Digital Question Paper Library Page

Figure 6.10 illustrates the Digital Question Paper Library where students can search and download previous year question papers.

The Digital Question Paper Library eliminates the need for students to search through physical cupboards or depend on photocopies. All papers are organized by department, subject, year, and semester, making it easy to find exactly what is needed. The search and filter system allows students to narrow down results quickly. Papers uploaded by administrators are immediately available to all students. The download button retrieves the PDF file from the server and opens it in the student's browser or downloads it to their device. This feature is particularly valuable during examination preparation periods when demand for past papers is highest.

ENTRY LOG AND OCCUPANCY DASHBOARD

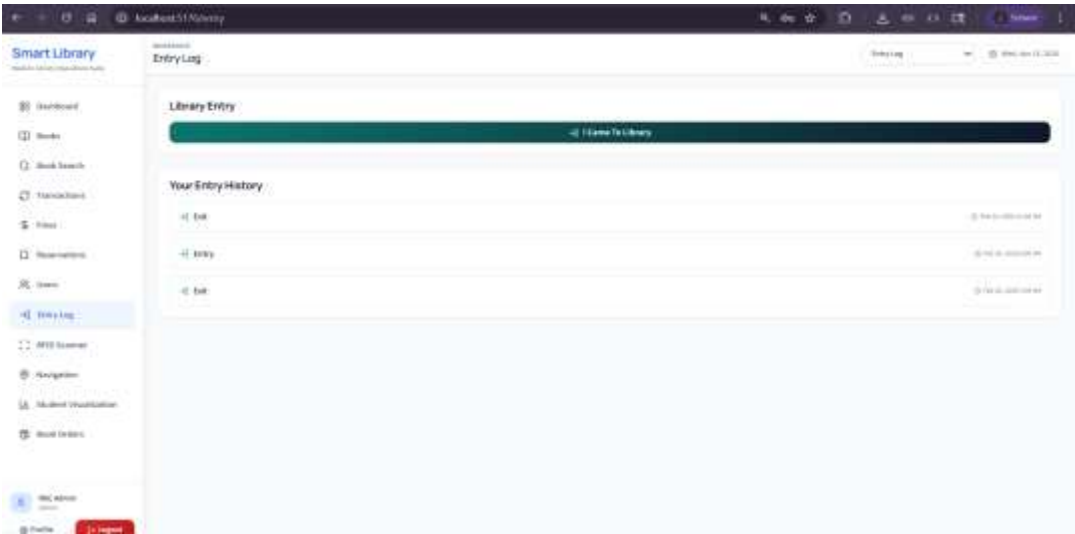


Fig 6.11 Entry Log and Library Occupancy Dashboard

Figure 6.11 illustrates the Entry Log and Library Occupancy Dashboard showing real-time student count and daily entry patterns.

The Entry Log and Occupancy Dashboard gives librarians and administrators unprecedented visibility into library usage patterns. The real-time occupancy counter at the top is updated automatically every time the hybrid GPS system logs a student entry or exit. The occupancy chart shows how the number of students inside the library changes throughout the day, clearly identifying peak hours. This data helps management decide when to schedule additional staff, when to open additional seating areas, and when to run maintenance activities with minimal disruption to students. The entry log table below provides a detailed audit trail of every student entry, including the confidence score from the hybrid GPS system and whether all three verification layers (GPS, Wi-Fi, motion) were satisfied.

CHAPTER 7

CONCLUSION AND SUMMARY

7.1 Conclusion

The Smart Library Automation System successfully addresses the challenges faced by traditional manual library systems by providing a unified, automated, and scalable solution built on modern web technologies and IoT hardware integration. The system integrates multiple library functionalities such as hybrid GPS-based automatic student entry, real-time RFID book shelf tracking, BLE beacon indoor navigation, digital question paper access, transaction management, fine calculation, and live analytics into a single platform. This integration eliminates manual registers, removes the problem of untrackable misplaced books, and provides students with an efficient, independent book-finding experience.

The core innovation of the system — real-time RFID shelf tracking — is a genuine advancement over existing library software. By dynamically updating a book's actual physical location every time a shelf scanner detects it, the system always knows where every book truly is, not just where it was originally shelved. Combined with BLE beacon indoor navigation, this creates a complete end-to-end solution where a student can search for a book from their phone and be guided physically to the exact shelf where it currently sits.

The system also emphasizes security and data integrity through role-based access control, JWT authentication, Google OAuth integration, and validation mechanisms. Automated processes such as entry detection, shelf location updates, fine calculation, and report generation significantly reduce librarian workload and improve accuracy.

The use of modern technologies including React, Node.js, Express, and MySQL ensures scalability, performance, and maintainability. The system is designed to handle large book collections and high student traffic while providing a responsive, user-friendly interface across all devices. In conclusion, the Smart Library Automation System provides a comprehensive and reliable solution for managing library operations in higher education

institutions. It enhances student experience, supports data-driven library management, and introduces real-time physical-digital integration that is not present in any conventional library management system.

7.2 Summary

This project presents the design and development of a full-stack Smart Library Automation System aimed at automating and streamlining all core library processes. The system begins with administrative configuration, where the library's physical layout — shelves, zones, RFID scanner positions, and BLE beacon placements — is registered in the system along with the complete book catalog with RFID tag assignments.

Students interact with the system beginning with automatic entry detection through the hybrid GPS, Wi-Fi, and motion sensor system, which logs their check-in without any manual action. When students search for a book, the system retrieves its real-time shelf location from the RFID tracking database. BLE beacons then guide the student step-by-step from their current position to the exact shelf. Librarians process borrowing and return transactions by scanning RFID tags, and the system automatically updates book locations as returned books are placed back on shelves.

The system calculates fines automatically for overdue books and provides a complete payment recording workflow. A digital question paper library allows students to search and download previous year papers without visiting physical storage. The reporting module generates comprehensive reports on entry patterns, borrowing activity, peak hours, and department-wise usage, which can be exported for management review.

Throughout the workflow, the system ensures data consistency, real-time synchronization between hardware and software, and strict security. Features such as role-based access control, confidence scoring in entry detection, dynamic RFID location updates, and modular architecture contribute to system reliability and correctness.

Overall, the Smart Library Automation System provides an efficient, transparent, and scalable solution for managing library activities. It reduces manual intervention, improves the accuracy of book location information, enhances the student library experience.